

Quantitative work: morning readout

10 September 2026 | Partial results | Status checked at 08:23 EDT

We have made substantial progress in solving the historical price path, but we do not yet have a completed new calibration or a matched policy result. The 100-date sequential path now reproduces exactly at its best prices. Its largest housing-market error is 0.04527%, still above our unchanged 0.02% requirement.

What was solved overnight

At the eleven inherited parameter values, the solver adjusted all 100 dated prices jointly. For each candidate price path, households anticipate the entire future from 2007; their decisions are solved backward and the population is propagated forward. A verified 100-column price-response matrix guides the price updates. This matrix concerns market clearing; it is not evidence that the economic parameters are identified.

Path evaluation	Largest market error	Meaning
1	18.199032%	Initial prices
2	6.984612%	Updated prices
3	1.914974%	Updated prices
4	0.223532%	Updated prices
5	0.045272%	Updated prices
6	0.045272%	Fresh final replay

All six evaluations passed their mapping, accounting and feasibility checks. The final replay has zero difference in prices and residuals; the target-fit, parameter, measurement and transition tables are byte-identical to the selected evaluation. This checks reproducibility, not convergence.

The remaining work, in order

- 1. Finish market clearing.** Continuation job 17319026 is running on Torch. At 08:23 it was in its first path, date 44 of 100, with a fresh heartbeat. The bounded round allows three paths including a final replay; there is no automatic further round.
- 2. Establish horizon stability.** The terminal unit-rent gap is 1.0867%, above its 1% gate. Other recorded endpoint checks pass. Historical prices and moments still need comparison across sufficiently long horizons; extending the endpoint alone is not a certificate.
- 3. Reconcile measurement and recalibrate.** Four pooled ACS targets are currently compared with a 2023 model cross-section, and two family-group definitions remain unresolved. Keep the reviewed childbirth regression unchanged. Then assess parameter sensitivities and conduct matched re-estimation before producing policies.

Fit at the reproduced provisional prices

Objective: 94.47633971. This is an inherited-parameter diagnostic, not a calibrated-equilibrium loss. Every target and weight is unchanged. Shares and their gaps are in fraction units.

Moment	Target	Model	Gap	Weight	Loss
Completed fertility	1.918	1.777103	-0.1408969	1425.739	28.3037
Childless share	0.188	0.2335966	0.04559661	17180.74	35.71964
Mean age at first birth	26.04463	26.23325	0.1886264	44.44444	1.581329
First births at age 30+ (share)	0.2603274	0.2440931	-0.01623427	10000	2.635514
First-birth housing response (rooms)	0.7202463	0.4113959	-0.3088504	137.5653	13.12215
Rooms gap: 3+ versus 1-2 children, ages 30-55	0.3676996	0.3474083	-0.0202913	2958.515	1.21813
Parent ownership gap (share units)	0.1676617	0.1465878	-0.0210739	14229.59	6.319493
Ownership share	0.575472	0.5875077	0.01203568	1207.846	0.1749657
Mean occupied rooms, ages 18-85	5.77997	6.32622	0.5462495	11.97316	3.572653
Wealth / annual gross labor earnings	6.8731	7.057288	0.1841884	6.287669	0.2133115
Annual bequests / wealth	0.0088	0.008354384	-0.0004456161	5165289	1.025691
Wealth / income dispersion, ages 76-84 (p90/p50)	3.448111	3.346356	-0.1017551	56.95977	0.5897673

What the fit tells us

Completed fertility is 1.777 versus 1.918, and childlessness is 23.36% versus 18.80%. These two rows contribute about 64.02 of the 94.48 objective. The first-birth housing response is 0.411 rooms against 0.720, while average occupied housing is 6.326 rooms against 5.780. Better market clearing alone has not repaired these economic discrepancies.

Which years are being targeted

The initial economy approximates a pre-2007 steady state, normalized to completed fertility 2.1. Households learn the path immediately in 2007. The model is measured along that transition: 2023 is an observation date, not a steady state. Cohort and wealth moments retain their approved source-specific dates; four ACS rows pool 2012-2023, and the birth-response branch uses 2019-2023. This objective does not fit an annual historical time series.

The historical age composition is imposed through 2023; the person-demographic model advances the population afterward. This remains a conditional historical fit. The numerical target fingerprint alone cannot resolve sample, family-group or timing differences.

Policy status

There is no new matched perfect-foresight property-tax comparison to report. The intended main comparison raises the annual tax from 1% to 2%, with equal household rebates in both regimes, starting from the same certified historical 2023 state. The unrebated historical baseline and the rebated policy baseline are distinct objects. Older policy effects must not be presented as results of this unfinished calibration.

Parameters and remaining checks

All eleven free coordinates are inherited inputs, not new estimates. Bounds and near-bound flags below reproduce the saved parameter contract. A target count of twelve against eleven coordinates does not itself establish identification.

Parameter	Value	Lower	Upper	Role	Near bound
beta_annual	0.9952766	0.94	0.9995	Free input	No
kappa_fert	2.168173	0.02	50	Free input	No
kappa_fert_continuation	1.770771	0.02	50	Free input	No
chi	1.053727	0.1	5	Free input	No
H0	13.71605	0.2	80	Free input	No
theta0	0.5703499	0	8	Free input	No
theta1	0.103724	0.02	16	Free input	Yes
hbar_child_rooms	0.2470849	0.1	1.8	Free input	No
first_birth_fixed_cost	4.619731	0	8	Free input	No
hbar_first_child_jump	0.4697651	0	0.5	Free input	No
psi_child_change_2023	-0.3213878	-1.5	0.2	Free input	No
psi_child_2007	0.2900515	-	-	Normalized	No
psi_child_2023	-0.03133627	-	-	Derived	No
tenure_choice_kappa	0.005	-	-	Fixed	No
housing_supply_elasticity	0.63	-	-	Fixed	No

Old fertility preferences are normalized to the initial completed-fertility target. The 2023 preference level equals that old intercept plus the transition coordinate. Tenure taste dispersion (0.005) and housing-supply elasticity (0.63) remain externally fixed. The near-bound flag concerns theta1; it is not a claim of parameter identification.

All endpoint-distance checks

Distance	Value	Tolerance	Pass
asset price relative gap	0.001544771	0.01	Yes
equal transfer relative gap	0	0.01	Yes
household heads relative gap	0.003217526	0.01	Yes
normalized head age sex l1	0.000722545	0.02	Yes
normalized household distribution l1	0.01526372	0.02	Yes
normalized person age sex l1	0.0008886332	0.02	Yes
psi absolute gap	0	0.001	Yes
renter price relative gap	0.01086682	0.01	No
resident persons relative gap	0.003168395	0.01	Yes

Evidence: HORIZON100_PROGRESS.md, horizon100_final_reproduction_review.json and overnight_target_mapping_review.md under output/model/e5f_matched_pf_20260909a/. The numerical snapshot is 96a41873; the original and continuation results remain separate. No source, target, weight or numerical gate changed for the continuation.